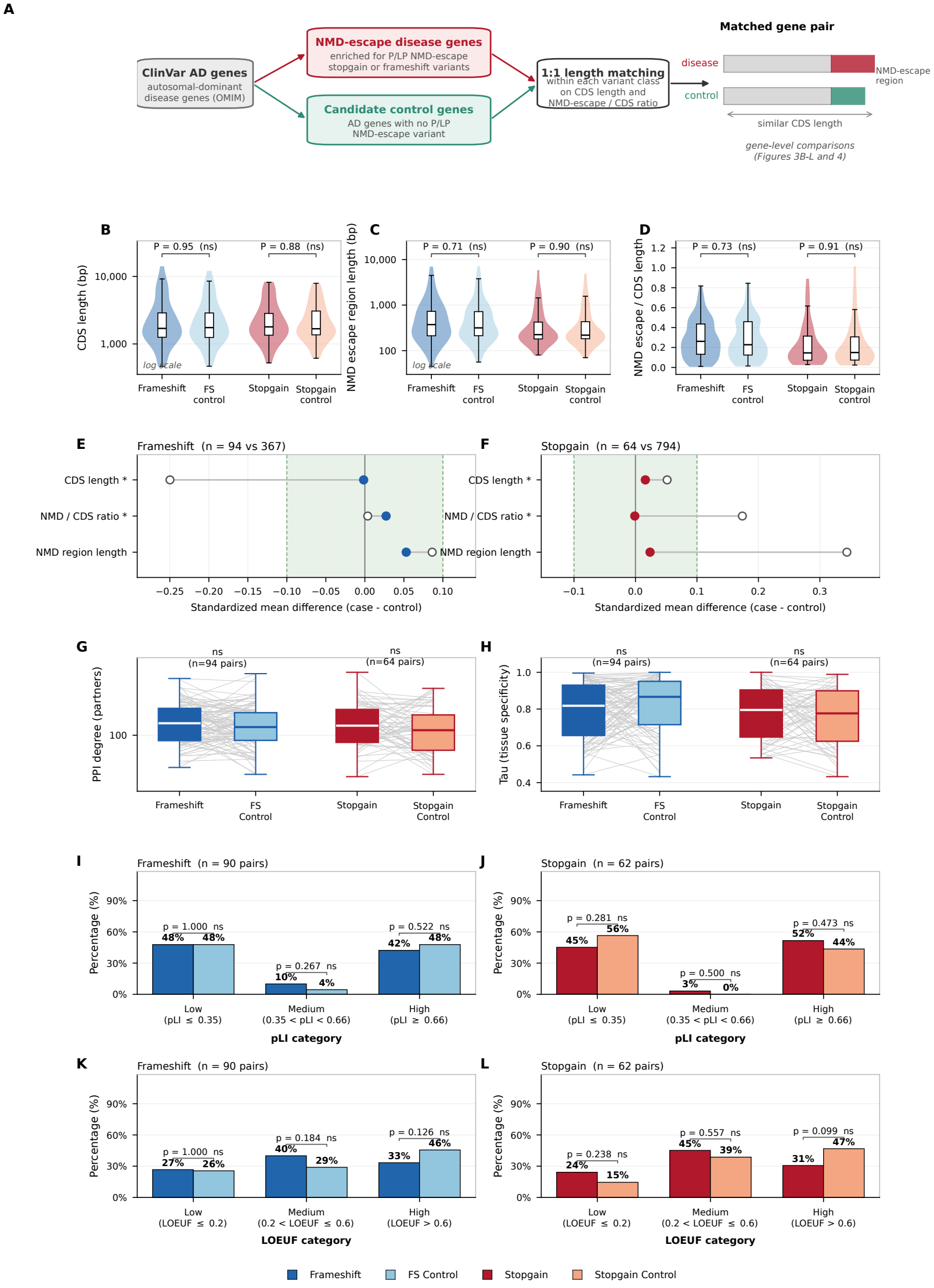

A
ClinVar AD genes
autosomal-dominant disease genes (OMIM)
NMD-escape disease genes
enriched for P/LP NMD-escape stopgain or frameshift variants
Candidate control genes
AD genes with no P/LP NMD-escape variant
1:1 length matching
within each variant class on CDS length and NMD-escape / CDS ratio
Matched gene pair
disease
control
NMD-escape region
similar CDS length
gene-level comparisons (Figures 3B-L and 4)
B
CDS length (bp)
P = 0.95 (ns)
P = 0.88 (ns)
log scale
Frameshift
FS control
Stopgain
Stopgain control
C
NMD escape region length (bp)
P = 0.71 (ns)
P = 0.90 (ns)
D
NMD escape / CDS length
P = 0.73 (ns)
P = 0.91 (ns)
E
Frameshift (n = 94 vs 367)
CDS length *
NMD / CDS ratio *
NMD region length
Standardized mean difference (case - control)
F
Stopgain (n = 64 vs 794)
G
PPI degree (partners)
ns (n=94 pairs)
ns (n=64 pairs)
FS Control
Stopgain Control
H
Tau (tissue specificity)
I
Frameshift (n = 90 pairs)
Percentage (%)
p = 1.000 ns
48%
48%
p = 0.267 ns
10%
4%
p = 0.522 ns
42%
48%
Low (pLI ≤ 0.35)
Medium (0.35 < pLI < 0.66)
High (pLI ≥ 0.66)
pLI category
J
Stopgain (n = 62 pairs)
p = 0.281 ns
45%
56%
p = 0.500 ns
3%
0%
p = 0.473 ns
52%
44%
K
Frameshift (n = 90 pairs)
p = 1.000 ns
27%
26%
p = 0.184 ns
40%
29%
p = 0.126 ns
33%
46%
Low (LOEUF ≤ 0.2)
Medium (0.2 < LOEUF ≤ 0.6)
High (LOEUF > 0.6)
LOEUF category
L
Stopgain (n = 62 pairs)
p = 0.238 ns
24%
15%
p = 0.557 ns
45%
39%
p = 0.099 ns
31%
47%
Frameshift
FS Control
Stopgain
Stopgain Control